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Hyper-spectral photonic integrated circuits based on recursive inverse design

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Manipulating optical spectra with complex spectral responses is central to high-performance photonic computing and sensing, but conventional integrated circuits typically increase features by cascading devices, which scales complexity linearly with cascade depth and is difficult to expand. Here, we introduce hyper-spectral photonic integrated circuits comprising a chain of interconnected reflective nodes. The central idea is to introduce partial reflectivity, thereby leveraging both transmissive and reflective interference within each component node. This architecture expands the accessible delay distribution and exhibits a superlinear complexity growth. We implement the circuits through a recursive inverse-design workflow that jointly exploits node geometry and circuit topology to maximize spectral complexity. Experimentally, a single waveguide incorporating 11 reflective nodes achieves a complexity comparable to at least 100 cascaded Mach-Zehnder interferometer stages. An array of these circuits reconstructs spectra with sub-picometer resolution across an 800 nm bandwidth. This performance bridges optical and microwave frequencies by enabling simultaneous monitoring of optical and radio signals within a single device.

Spectral complexity is a key metric for optical systems, measuring the diversity of independent spectral features present in frequency responses. This parameter directly governs the achievable density of spectral channels, and in turn, the computational throughput, multiplexing capability, and resolution attainable in applications such as optical computing[1], sensing[2], and signal processing[3,4]. While conventional free-space optics often requires bulky setups to achieve high spectral manipulation precision, integrated photonics provides a compact and scalable platform for realizing complex photonic integrated circuits (PICs). Over the past two decades, substantial progress has been made in this direction. For example, in communications, high-density spectral multiplexing in integrated photonics has enabled optical interconnect densities exceeding 5.3 Tbps mm^{-2}[5–7]. Similarly, parallel processing of multiple wavelengths on-chip has facilitated photonic computing densities beyond 0.1 TOPS mm^{-2}[8,9]. In sensing, the availability of numerous independent spectral channels has boosted the development of computational spectroscopy[10,11], with integrated spectrometers now rivaling the performance of their benchtop counterparts.

Despite these advances, current approaches for scaling up the spectral complexity of PICs remain rudimentary. In most implementations, the complexity is increased by cascading discrete photonic components such as Mach-Zehnder interferometers, ring resonators, and Fabry-Perot filters, where the overall spectral response results from the simple superposition of individual contributions[12–14]. As a result, the total spectral complexity scales only linearly with the number of components. As the cascade depth increases, the number of elements, optical loss, and calibration overhead grow substantially,

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while the improvements in spectral structure become marginal. To overcome this limitation, several strategies have been proposed in recent years, including the use of disordered photonic media[15,16], exploitation of chaotic dynamics[17,18], and advanced tuning schemes[4,19]. Although these methods can improve system performance to some extent, they often depend on stochastic or intuitive design paradigms, which constrain their operating bandwidth and the scope of complexity. A comprehensive comparison of the existing methods on complex response generation is provided in Supplementary Table 3.

Here, we introduce an on-chip architecture termed Hyper-Spectral Photonic Integrated Circuits (HS-PICs) for generating high-complexity spectral responses with super-linear scaling. It consists of a chain of connected reflective nodes, each of which distributes incident light into forward and backward directions. In contrast to conventional counterparts that rely on a single interference modality, these circuits scale complexity by combining two distinct mechanisms: transmissive and reflective interference, thereby expanding spectral control to a wider complexity range. The realization of such circuits is enabled by our development of recursive inverse design, a hierarchical workflow that performs hyper-spectral design in an end-to-end fashion. This approach operates by recursively inserting new reflective nodes into an initial optical link and optimizing local regions. By modeling both device geometry and inter-node topology as design degrees of freedom, this method accesses a substantially larger design space for high complexity responses.

Using this strategy, we show that even a single waveguide incorporating only 11 inversely designed reflective nodes can achieve a hyper-spectral complexity at least ten times finer than that of MZI architecture. A spectrometer comprising an array of such circuits achieves sub-picometer resolution over an 800 nm bandwidth, in both cases limited only by our measurement setup. These HS-PICs enable applications that require dense spectral encoding. We demonstrate that their sub-picometer spectral resolution bridges the conventional divide between optical and microwave domains: the same chip reconstructs both optical spectra and microwave signals, with 125 MHz radiofrequency resolution across more than 20 THz of optical bandwidth. We apply this capability to monitor signals from a microwave photonic radar and high-speed optical communication links. Owing to their high spectral complexity, HS-PICs can support a broad range of applications in integrated photonics.

Results

The concept and principle of hyper-spectral photonic integrated circuits

To describe the complexity of a spectral response, we establish a theoretical framework for dispersive optical systems by connecting established temporal and frequency domain definitions. For a linear time-invariant system, $h(\tau)$ is the impulse response and

$$P(\tau) = |h(\tau)|^2, \qquad p(\tau) = \frac{P(\tau)}{\int P(\tau)\, d\tau}, \tag{1}$$

are the power delay profile (PDP) and its normalized form, respectively; $P(\tau)d\tau$ is the output energy arriving within $[\tau, \tau + d\tau]$ [20]. The complex frequency response is $H(\omega) = \int h(\tau)e^{-i\omega\tau}d\tau$, and the normalized field correlation is

$$\Gamma(\Delta\omega) = \frac{\left\langle H(\omega)H^*(\omega+\Delta\omega)\right\rangle_\omega}{\left\langle |H(\omega)|^2\right\rangle_\omega} = \int p(\tau)e^{i\Delta\omega\tau}d\tau \tag{2}$$

which forms a Fourier-transform pair with $p(\tau)$. Here, $\langle\cdots\rangle_\omega$ denotes averaging over the analyzed spectral interval. Previous speckle and computational spectrometers commonly characterize the experimentally measured intensity spectrum $I(\omega) = |H(\omega)|^2$ using the normalized intensity correlation[15,21]

$$R(\Delta\omega) = \frac{\left\langle I(\omega)I(\omega+\Delta\omega)\right\rangle_\omega}{\left\langle I(\omega)\right\rangle_\omega\left\langle I(\omega+\Delta\omega)\right\rangle_\omega} - 1 \tag{3}$$

The Siegert relation gives $R(\Delta\omega) = |\Gamma(\Delta\omega)|^2$. Thus, $R(\Delta\omega)$ is an experimentally accessible representation of the PDP–field-correlation Fourier relation, with a broader temporal response yielding a more rapidly decaying spectral correlation.

Starting from this correlation function $R(\Delta\omega)$, we introduce the spectral complexity C as a more direct scalar measure of response complexity. We first define the decorrelation width $\Delta\omega_\alpha$ as the smallest positive offset satisfying

$$R(\Delta\omega_\alpha) = \alpha R(0), \tag{4}$$

and then define

$$C_\alpha = \frac{1}{\Delta\omega_\alpha}. \tag{5}$$

When using $\alpha = 0.5$, $\Delta\omega_{0.5}$ becomes the half-width at half-maximum of $R(\Delta\omega)$. A larger C therefore directly denotes a more rapidly decorrelating and more complex spectral response. For wavelength-sampled spectra, the equivalent quantity is $C = 1/\Delta\lambda_{0.5}$ in nm^{-1}, where locally $\Delta\omega \simeq (2\pi c/\lambda_0^2)\Delta\lambda$. We further show that a mathematical relationship between this metric and the normalized PDP $p(\tau)$ exists. By performing a Taylor series expansion, the PDP–correlation Fourier relation gives:

$$C_\alpha \simeq \frac{\sigma_\tau}{\sqrt{2(1-\sqrt{\alpha})}}, \quad \sigma_\tau^2 = \int (\tau-\bar{\tau})^2 p(\tau)d\tau. \tag{6}$$

where $\bar{\tau}$ and σ_τ are the expectation and root-mean-square (RMS) of $p(\tau)$. Therefore, for a fixed correlation criterion α, spectral complexity is proportional to the RMS delay spread and differs only by a constant coefficient. This shows that a complex spectral response is physically equivalent to a temporally dispersed impulse response. The complete derivation is provided in Supplementary Note 1.

These definitions enable an intuitive analysis of PIC complexity through $P(\tau)$. We identify two primary mechanisms for generating delay differences $\Delta\tau$. The first employs transmissive interferometry, such as Mach-Zehnder Interferometers (MZIs) or Arrayed Waveguide Gratings (AWGs), to disperse optical power by introducing arm length differences (Fig. 1a, top). These delays result in a bell-shaped $P(\tau)$ distribution (Fig. 1a, bottom). Since the extrema of τ are bounded by the maximum and minimum path lengths, increasing the variance σ_τ necessitates a higher cascade count. Our analytical model for unbalanced MZIs confirms that this architecture follows a linear complexity scaling relative to the cascade depth:

$$C_{\mathrm{MZI}}(N) = \mathcal{O}(N) \tag{7}$$

where $\mathcal{O}(N)$ represents the linear dependence on cascade depth N (refer to Supplementary Note 2). Alternatively, reflective architectures like Bragg Gratings or stratified waveguides create optical path differences by introducing back-propagation (Fig. 1b, top). The resulting temporal profile consists of a primary energy peak representing unreflected light, followed by a series of trailing reflection peaks (Fig. 1b, bottom). The delay $\Delta\tau$ from the main peak is determined by the back-propagation depth. While increasing the number of reflectors enhances the variance, it inevitably leads to a significant reduction in system transmittance. Previous studies have demonstrated that a 30 μm stratified waveguide can suffer losses exceeding 20 dB, thereby making further complexity improvements difficult[22]. These time-

domain behaviors discussed have been verified through simulations (Supplementary Note 4).

Here, we propose the HS-PIC as a dispersive photonic architecture that combines transmissive and reflection-induced delay mechanisms. The HS-PIC is defined as a delay waveguide chain interconnected by reflective nodes (RNs) (Fig. 1c). Herein, the waveguides provide arm differences, while reflectors generate counter-propagating optical path differences to maximize temporal energy delocalization. The $P(\tau)$ of HS-PIC reveals the synergy of two mechanisms: firstly, energy is stratified into distinct orders based on reflection depth. The n-th order corresponds to back-propagation energy going through n reflectors, introducing a delay denoted as $\Delta\tau_r$. Inside each order, the energy exhibits dispersion caused by transmissive interference, denoted as $\Delta\tau_t$. The aggregate delay is thus driven by the summation $\sigma_\tau = \sum_{m=1}^{M}\Delta\tau_{tm} + \sum_{n=1}^{N}\Delta\tau_{rn}$. Crucially, because multiple reflections allow the optical path order n to exceed the number of physical components N, the actual σ_τ surpasses this linear summation.

To illustrate this intuition, we first simulated randomly generated HS-PIC structures with increasing stage number and compared them with cascaded MZI and Bragg-grating baselines under the same stage-count coordinate (Fig. 1d). This comparison provides an architecture-level visualization of the observed super-linear complexity trend. We further benchmarked the cascaded MZI chain, coupled microring chain, chirped Bragg baseline, and HS-PIC under matched passive-area and insertion-loss constraints (Fig. 1e and Note 3). This resource-matched analysis shows distinct upper-bound scaling laws: $C_{\mathrm{MZI}} \sim A$, $C_{\mathrm{MRR}} \sim \sqrt{A/a_{\mathrm{MRR}}}$, $C_{\mathrm{Bragg}} \sim L$, and $C_{\mathrm{HS}} \sim A^{1.11}$ within the investigated design space. At $A = 10^5\ \mu\mathrm{m}^2$ and $L = 20$ dB, the HS-PIC has at least 8.19 times higher complexity upper bound than all three baselines. Thus, the super-linear behavior is presented as a constraint-dependent empirical trend of the HS-PIC layouts.

Recursive inverse design for hyper-spectral engineering

The HS-PIC requires the co-design of component geometries and circuit-level parameters, whereas conventional methodologies have typically focused on only one of these aspects at a time. Circuit synthesis methods such as Layer-Peeling Synthesis (LPS) can efficiently match a target spectrum, but return idealized coupling profiles that do not guarantee a fabrication-compatible geometry [23]. Similarly, Full-wave topology optimization incorporates device physics and fabrication constraints, but end-to-end optimization of a millimeter-scale circuit is computationally prohibitive[24–26]. At its core, the HS-PIC design problem requires an algorithm that bridges geometry and circuit design.

Inspired by the recursive concept in computer programming, we developed a recursive inverse design algorithm to implement the process-friendly full circuit design for high complexity targets. In computer science, recursion decomposes a complex task into smaller, self-similar subproblems that can be efficiently solved through repeated application of the same operation. The corresponding computational cost follows $T(n) = aT\left(\frac{n}{b}\right) + \mathcal{O}(n^c)$, where $a < b^c$ ensures sublinear growth of $T(n)$ relative to direct computation[27]. Analogously, our framework applies this principle to photonics by partitioning the high-dimensional design parameter space into tractable subproblems. The process comprises three stages: decomposition, base-case solving, and unwinding. In the decomposition stage, each recursive step inserts a single reflective node to the current circuit, jointly optimizing its

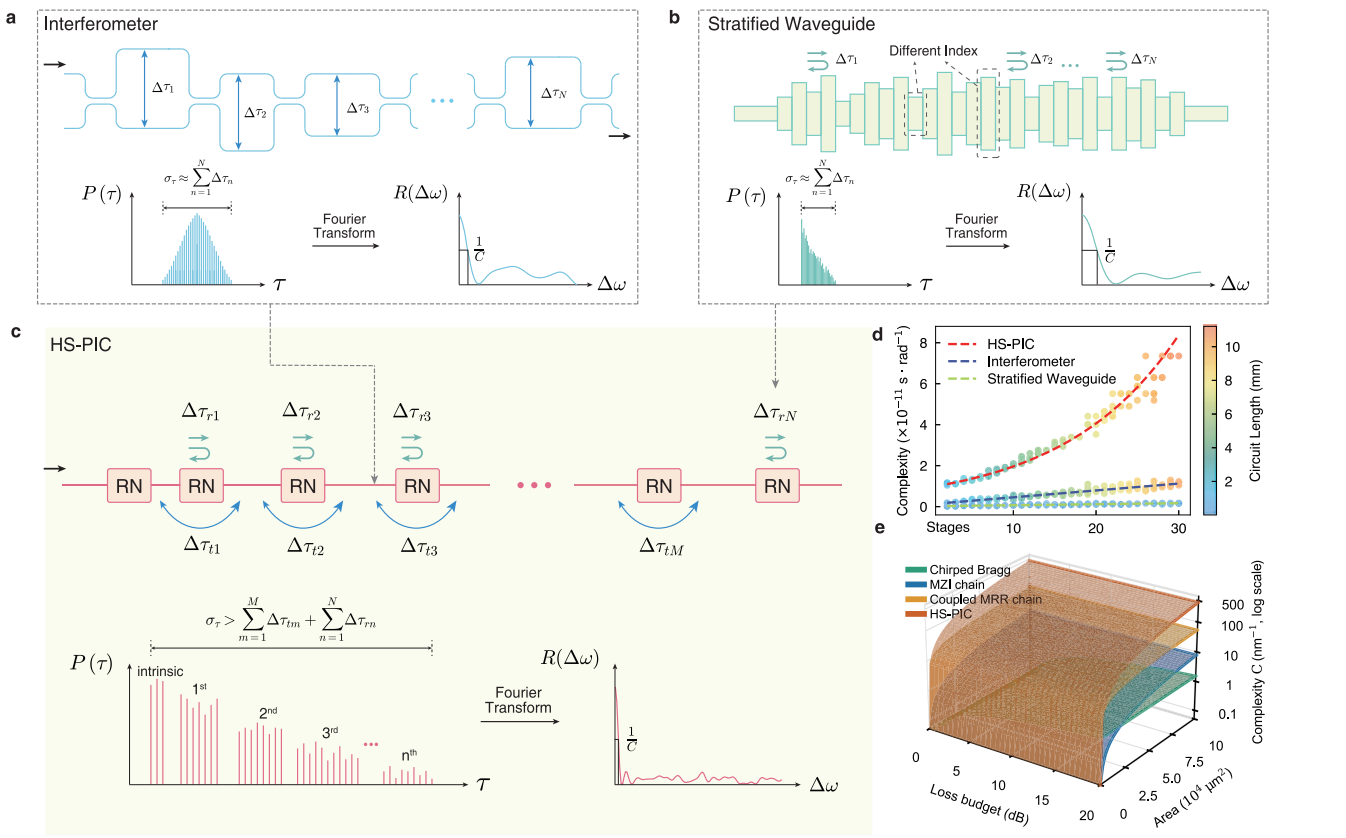


Fig. 1 | Concept and principle of hyper-spectral circuit. a–c Circuit structures and complexity analysis. The Fourier transform relationship between the system power delay function (left) and the frequency auto-correlation (right) elucidates the physical origin of the spectral complexity. **a** A cascaded interferometers scales complexity via physical path differences. **b** Waveguide Bragg gratings scale complexity through reflection-induced path differences. **c** The HS-PIC combines both mechanisms. **d** Illustrative complexity scaling versus stage number for representative architectures. The dashed lines represent linear and power-law fits. **e** Upper-bound complexity comparison under matched passive-area and insertion-loss constraints for cascaded MZI chains, coupled microring chains, chirped Bragg filters, and HS-PICs.

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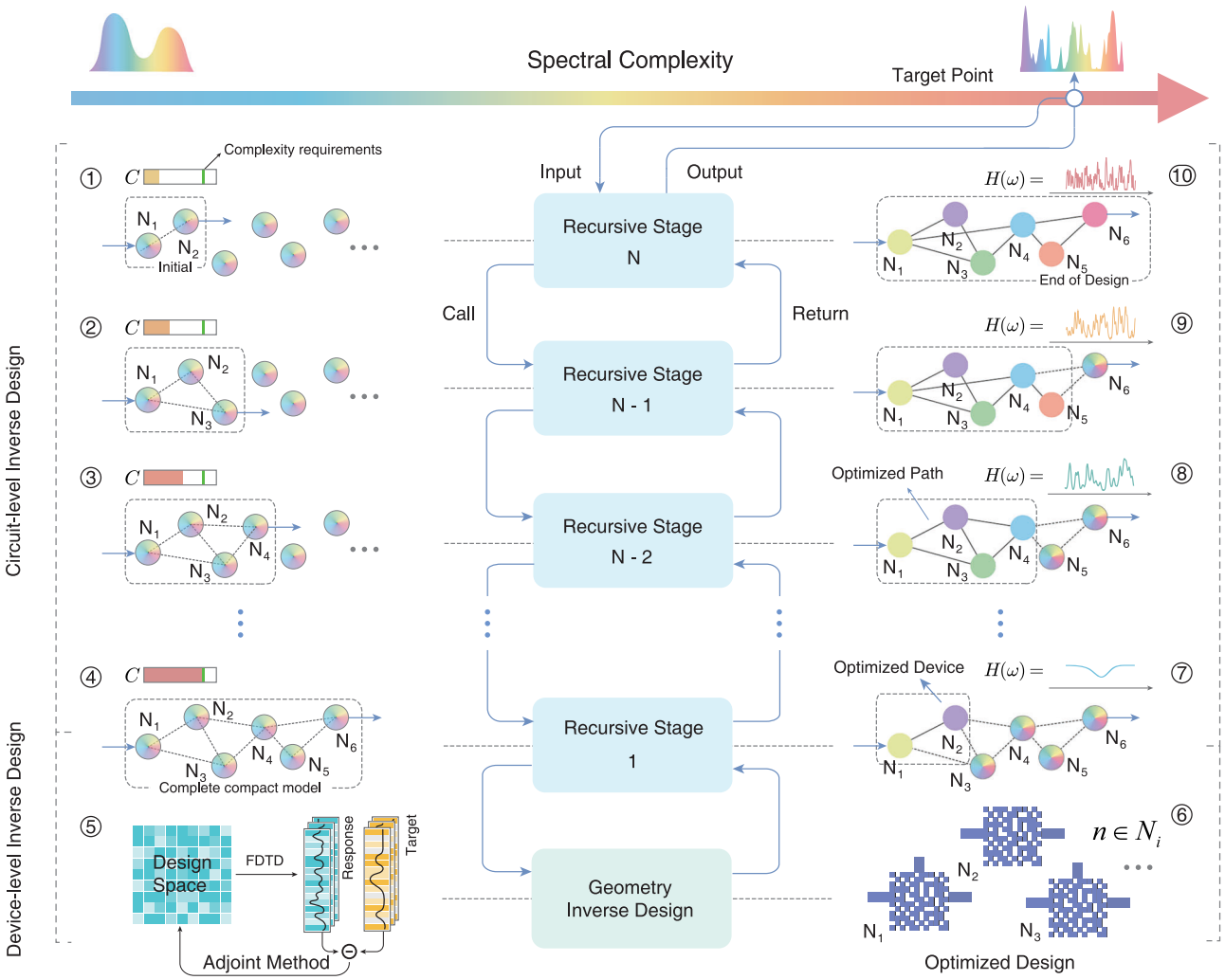


Fig. 2 | Recursive inverse design enables high-complexity spectral synthesis. Schematic of the recursive inverse design (RID) framework. The algorithm takes the target spectrum complexity C as input and outputs the complete device geometry and circuit layout, including the number of reflective nodes N at which the recursion converges. The top color bar denotes increasing C and spectral features from left to right. The workflow consists of a sequential process marked 1 to 10: labels 1–4 are the decomposition stage, during which a single reflective node is added per iteration until the achieved complexity meets C; labels 5 to 6 are the geometry optimization stage; and labels 7–10 are the unwinding stage that replaces the compact model into fabricatable devices.

insertion position, coupling coefficients, and adjacent waveguide lengths. The achieved complexity C_n is evaluated after each insertion, and the recursion terminates automatically once C_n first meets the user-specified target C. Consequently, the number of reflective nodes N is an emergent output of the algorithm rather than a manually chosen hyperparameter (Fig. 2a, left). Once the target is met and recursion terminates, compact models for all devices and their connections are obtained. Subsequently, geometry inverse design with fabrication constraints is performed under the guidance of compact models. This phase involves parallel running optimizations where each routine realizes a distinct reflector (Fig. 2a, bottom). During unwinding, the ideal compact models are sequentially replaced by their physically realizable device models, and the inter-device waveguide lengths are fine-tuned to compensate for model mismatch (Fig. 2a, right). Both the decomposition and unwinding stages are compatible with various algorithms and figures of merit (FoM). In this work, we used transfer-matrix models (TMM) and particle swarm optimization (PSO), with the objective function defined as:

$$\mathrm{FoM} = \Delta\omega_{0.5} + \beta \int_{\lambda} R(\Delta\lambda) \tag{8}$$

Here, the first term quantifies the spectral decorrelation distance, while the second term represents the integrated spectral correlation weighted by β. Minimizing this FoM promotes greater response complexity. (see "Methods" for methodological details on RID). Additional design targets beyond complexity control are presented in Supplementary Note 5.

We applied RID across 19 wavelength bands and five complexity levels, demonstrating complex spectral-design capability from the visible to the mid-infrared. At the lowest target complexity, C = 0.01 nm^{-1}, a single reflective node is sufficient, and the recursion terminates at N = 1. Here, geometry inverse design produces a broadband RN with near-zero intrinsic loss (Fig. 3a; see Supplementary Figs. 11, 12 for the complete FDTD field distributions). Under the same footprint constraint, the inverse-designed RN provides a broader usable bandwidth and lower insertion loss than a conventional Bragg reflector. Its fabrication-compatible minimum feature size is wavelength dependent, increasing from 150 nm at the shortest design wavelengths to 350 nm at longer wavelengths (Supplementary Note 10). As the target complexity increases from 0.1 to 100 nm^{-1}, RID automatically generates cascaded RN chains with recursion converged to N = 3, 7, 11, and 16 (Fig. 3b–e). With no more than 16 cascaded RNs, the HS-PIC reaches a complexity of 100 nm^{-1}, whereas a conventional

architecture would require substantially more stages under comparable resource constraints. The complexity evolution during recursion further reproduces the observed super-linear trend of the HS-PIC within the investigated design space (Fig. 3f). In total, the design sweep produced 19 × 5 = 95 sets of structural parameters that map directly to fabrication-compatible chip layouts (see Supplementary Tables 4, 5 for complete design parameters).

The RID allows convenient control over the optical loss through setting reflectivity constraints on components. The relation between loss and reflection coefficient is characterized by parameter sweeping as shown in Fig. 3g. By applying appropriate limits on reflectivity, the transmission power of all designs has been maintained above −5 dB (colored data points in the plot show the corresponding waveguide loss in A–E).

The recursive formulation improves the practical scalability of HS-PIC design by localizing full-wave optimization to compact reflective nodes and treating the long inter-device links with low-dimensional TMMs. Under a unified large-area 3D-FDTD benchmark, RID substantially reduces runtime and memory requirements compared with full-domain, blockwise, and pixelwise inverse-design workflows (Fig. 3h, Supplementary Note 6). The RID design presented here is completed in approximately 25 h on a dual-Xeon workstation, whereas the other workflows require weeks to years of serial computation time. In comparison, LPS is numerically inexpensive but requires an impractically large number of weak longitudinal reflector layers at high complexity, limiting its physical realizability. Meanwhile, approximately 95% of the RID optical path remains a low-loss straight waveguide, avoiding both the accumulated scattering of a long fully optimized region and the stitching loss of block-assembled designs. Thus, for the chain-structured synthesis problem studied here, RID provides a favorable combined trade-off between computational overhead, fabrication compatibility, and insertion loss.

Hyper-resolution spectrometer based on HS-PICs

HS-PIC enables ultrahigh-resolution spectral sensing. Figure 4c outlines the operating principle when a thermally tuned reflective node network is used as a spectrometer. As discussed in Fig. 1, the spectrum modulation at the output comes from the interference of all optical paths P_1 to P_n. Thermal tuning of the waveguides introduces the effective index change Δn, which produces controlled phase separation. Heating combinations across segments, therefore, set the relative phases of paths and produce a family of distinct spectral responses (Fig. 4d). Experimentally, we integrated thermo-optic phase shifters onto each delay waveguide segment of the fabricated HS-PIC (see "Methods" for chip fabrication). Each heater was driven by preset discrete voltage states (0 V and 5 V), which were experimentally confirmed to exhibit sufficiently low pairwise spectral correlation (see Supplementary Note 8 for coherence analysis). By increasing the preset voltage levels to three, this setup enabled the realization of an 11-stage HS-PIC spectrometer capable of providing 59,049 accessible tuning states(Supplementary Fig. 14).

The setup of the on-chip spectrometer is presented in Fig. 4a and the "Methods" section. The system is composed of a silicon nitride HS-PIC chip fabricated on a 4-inch, 300 nm silicon nitride wafer, optical edge couplers, a direct current (DC) probe card, and a high-speed driving board with a microcontroller unit (MCU, Supplementary Fig. 18). The chip distributes the optical signal under test through three Y-branches to four HS-PICs with $C = 10\ \mathrm{nm}^{-1}$, targeting bands centered at 780, 1310, 1550, and 2000 nm. The HS-PICs are arranged horizontally, with heaters in the same column connected in parallel for joint control (Fig. 4b). This configuration enables simultaneous spectral detection and allows the customized operating bandwidth. The chip exhibits total insertion losses of 16 dB, 14 dB, 11 dB, and 13 dB across the four ports under corresponding ASE illumination, while effectively suppressing out-of-band light. Although the loss of individual HS-PIC could not be measured directly, the insertion loss measurements imply a per-channel loss of less than 10 dB. This performance aligns well with numerical simulations. The calibration process was performed using three ASE sources spanning a total spectral coverage of 235 nm. Although the measured bandwidth was limited by the calibration sources, the symmetry of the response envelope indicates that the maximum operational bandwidth exceeds 400 nm (Supplementary Fig. 15). Analysis of the 1024 calibrated response channels gives average measured spectral complexities of C = 15.8, 11.5, and 6.80 nm^{-1} around 1310, 1550, and 1980 nm, respectively. These values are in agreement with the RID-optimized value of 9.75 nm^{-1}, confirming that the designed response complexity is retained experimentally. The maximum power consumption of each channel is experimentally measured at around 49 mW, resulting in an average spectrometer power consumption of 245 mW. The HS-PIC chip has a fast channel switching time below 0.5 ms (Supplementary Fig. 17), supporting real-time spectral acquisition. In its normal operating mode with 1024 channels, the HS-PIC performed a scan in 1 s, with channel count and acquisition time increasing under low-SNR conditions.

To characterize the spectrometer performance, we performed spectral reconstruction of single-peak, dual-peak, and wide-band lasers. The spectral encoding of our HS-PIC spectrometer represents an underdetermined system, which we solved using a least mean squares (LMS) method with regularization constraints. The norm terms in our recovery algorithm are expressed as:

$$\min_{0<\boldsymbol{\phi}<1} \| \mathbf{I} - \mathbf{T}\boldsymbol{\phi}\|_2 + \alpha \| \boldsymbol{\phi}\|_1 + \beta \| \boldsymbol{\phi}\|_2 + \gamma \| \mathbf{D}\boldsymbol{\phi}\|_1 \tag{9}$$

where α, β, γ denote the coefficients of the L_1, L_2 and total variation (TV) norms[28]. To quantify errors in the signal acquisition and processing, we defined the signal-to-noise ratio (SNR) of the detected signal and the spectral recovery accuracy (ε), which are given by $\mathrm{SNR}_{\mathrm{dB}} = 10\log_{10}(\| \mathbf{I}\|_2^2 / \| \mathbf{I} - \mathbf{I}_0\|_2^2)$ and $\varepsilon = \|\boldsymbol{\phi}_0 - \boldsymbol{\phi}\|_2 / \|\boldsymbol{\phi}_0\|_2$, respectively. The I and I_0 represent the detected and ideal intensity response. Results for single-peak lasers (Fig. 4e) demonstrate excellent uniformity and accuracy across an operational bandwidth spanning 800 nm ($\varepsilon < 0.05$). Figure 4f presents the spectral results for dual-peak lasers with various frequency spacings. For larger spacings (4 pm to 40 m), the input light was generated by two independent lasers, while for small spacings (1.75 pm and 0.9 pm), it was produced by a laser modulated by one or two 100 MHz acousto-optic modulators (AOM). This modulation approach is necessary to generate a stable dual-peak signal in the presence of approximately 10 MHz laser frequency noise. We believe that the resolution might be pushed further with a more stable laser. For wide-band light (Fig. 4g), while the signal SNR was reduced compared with dual-peak lights, the high channel number of 4096 provided sufficient redundancy to maintain a low recovery error ($\varepsilon < 0.1$). We also demonstrated robust spectral reconstruction under increased measurement noise and degraded fabrication fidelity. Stable reconstruction was maintained at a noise level of 400% and when the fabricated feature sizes did not meet the design specifications (see Supplementary Note 8).

The performance of the spectrometer is primarily limited by experimental noise sources. Specifically, temperature drift induced by the phase shifters compromises the ultimate spectral resolution. We experimentally verify the chip having a thermal sensitivity of 0.014 pm/mK (see Supplementary Note 7 for experimental results and analysis). To mitigate this, we reordered the thermal tuning sequence to minimize power fluctuations between adjacent phase shifters. This strategy successfully reduced the on-chip temperature drift by an order of magnitude to less than 10 mK, corresponding to a spectral accuracy of 0.14 pm. These results establish the engineering foundation for our high-precision, sub-picometer resolution. Second, HS-PIC spectrometers demonstrate strong robustness against power fluctuations. The unpackaged in-lab testing induces about 1 dB error on

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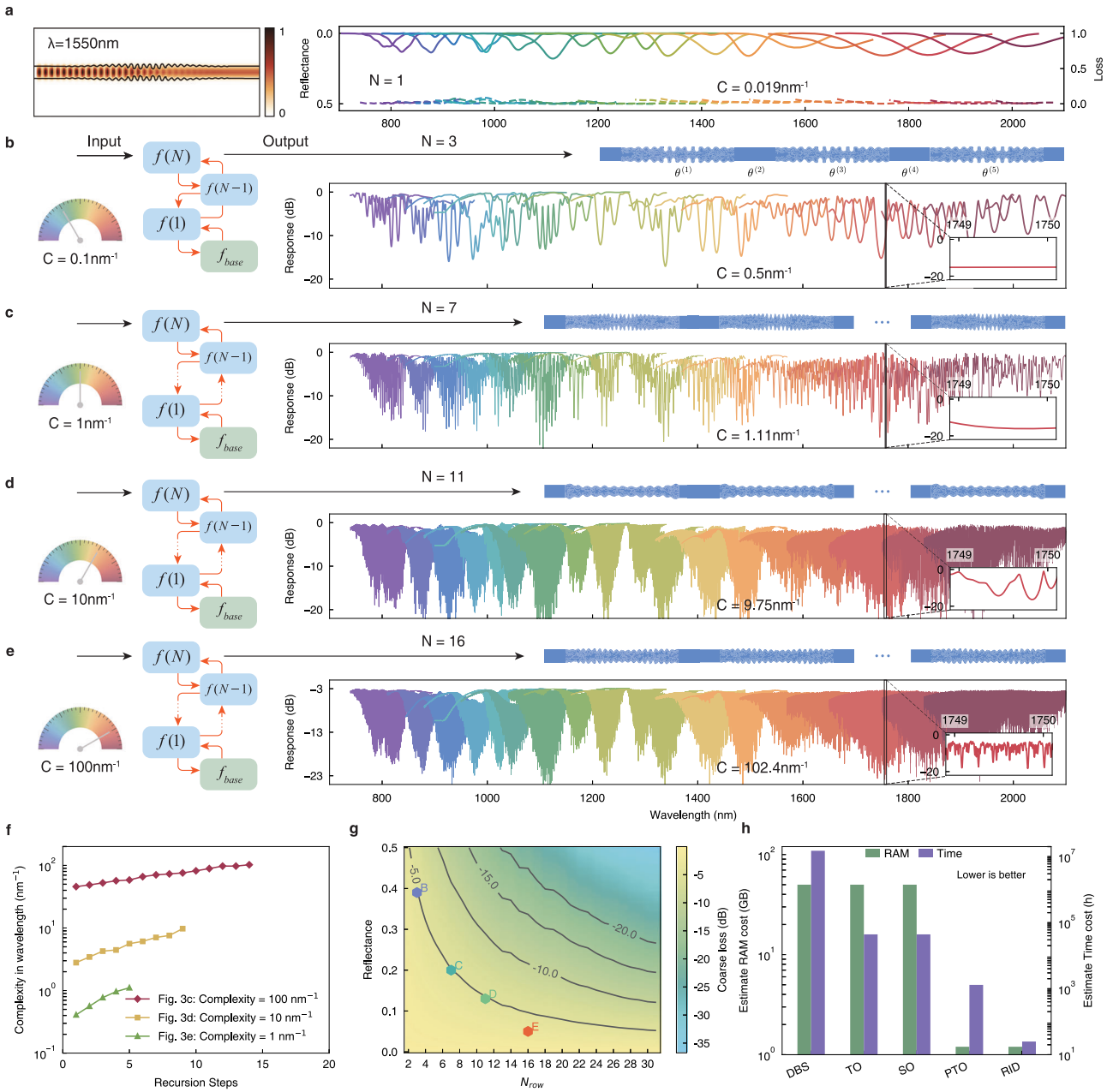


Fig. 3 | RID of HS-PICs across five target complexity levels. a A representative FDTD field distribution of a reflective node at 1550 nm and the simulated reflectance (solid) and loss (dashed) of the single-node designs across the investigated wavelength range. The single-node circuit achieves $C = 0.019$ nm^{-1} with $N = 1$. **b**–**e** RID results for target complexities of 0.1, 1, 10, and 100 nm^{-1}, respectively. Each panel shows the recursive construction, converged node count, physical RN-chain layout, and designed spectral response; Insets magnify a representative 1-nm spectral interval. **f** The evolution of spectral complexity with the recursion step for representative targets. **g** Estimated coarse loss as a function of RN reflectance and stage number N_{row}; markers B–E denote the designs in panels (**b**-**e**). **h** Computational benchmark of representative large-area inverse-design workflows for a common 2 × 1000 μm² waveguide region, comparing peak RAM and estimated serial wall-time of representative large-area inverse-design workflows under the unified 3D-FDTD benchmark; lower values indicate better computational efficiency.

intensity response, proven by low SNR in most experiments. However, the high number of scanning channels provides sufficient redundant information, allowing signal recovery with good fidelity even at a low SNR ratio. This resilience also suggests the spectrometer's potential for robust operation in low-light environments.

Unified microwave-optical spectrum analysis

The hyper-resolution accuracy of our approach expands the boundaries of on-chip measurement. As recent advances are summarized in Fig. 5e, integrated spectrometers work in several hundred nanometers bandwidth yet are constrained to near 10 pm maximum resolutions[10,15,28–53], while on-chip microwave frequency analyzers attain a high resolution of about 100 MHz within only about 1 nm of equivalent optical bandwidth[54–68]. Using the same spectrometer chip, we demonstrate a microwave spectrum analyzer that bridges this performance gap, achieving both wide bandwidth and high resolution on the same platform. As the experiment setup in Fig. 5a, left panel, the signal-under-test (SUT) is loaded using a carrier-suppressed single-sideband (CS-SSB) lithium niobate (LN) modulator, then encoded by our chip encoder. The reconstruction process of the microwave follows the steps in Fig. 5a, right panel. Because of the presence of a strong reference carrier, the retrieval algorithm is adapted to a two-

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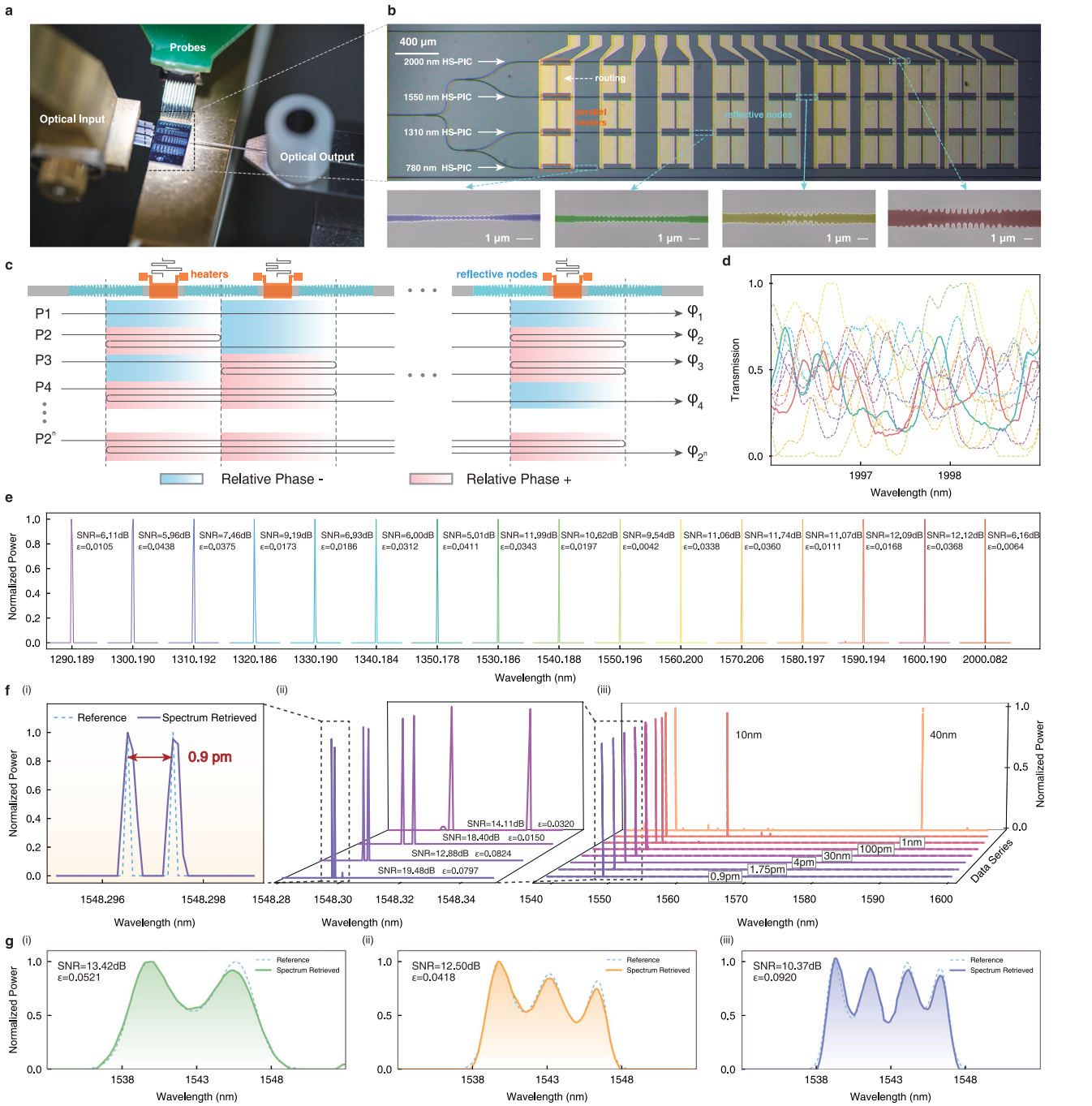


Fig. 4 | Setup and testing of the ultra-resolution spectrometer. a Testing setup of the spectrometer (left), **b** Optical charge-coupled device (CCD) image of the fabricated chip (upper), and false-color scanning electron microscope (SEM) images of the reflective nodes designed for different operating bands (lower). **c** The working principle of the spectrometer is where phase shifters modulate the relative phase of optical pathways. **d** Sample channels of the spectrometer, dashed lines are randomly selected channels, and the solid lines are the two closest channels, showing a distinct response. **e**–**g** Reconstruction result of (**e**) Single-peak laser (**f**) dual-peak laser, and (**g**) wide-band laser. The noise and performance are marked by SNR and ε.

step LMS procedure, allowing for more accurate recovery of both the carrier and signal, with the two norms separately written as:

$$\min_{p>0,k\in\Lambda} \| \mathbf{I} - \mathbf{T}p\mathbf{e}_k \|_2 \tag{10}$$

$$\min_{0<\boldsymbol{\phi}<1} \| \mathbf{I} - \mathbf{T}(\boldsymbol{\phi} + p\mathbf{e}_k) \|_2 + \alpha \| \boldsymbol{\phi} \|_1 + \beta \| \boldsymbol{\phi} \|_2 + \gamma \| \mathbf{D}\boldsymbol{\phi} \|_1 \tag{11}$$

where $\mathbf{T}$ is the transmission matrix from calibration. $p\mathbf{e}_k$ and $\boldsymbol{\phi}$ denote the carrier and signal within the reconstruction window Λ, respectively. To verify the extensive bandwidth capabilities of our design, we demonstrate parallel signal reconstruction across multiple carriers. We simultaneously inject signals for frequency locking, high-speed communication, and LiDAR, each utilizing a 30 GHz frequency range. As shown in Fig. 5b–d, multi-tone, QPSK, and FMCW signals are recovered with a resolution of 125 MHz. The average frequency mismatch is 157 MHz, calculated as the mean error across center and edge frequencies. This value is slightly larger than the intrinsic 0.9 pm spectrometer resolution, a discrepancy

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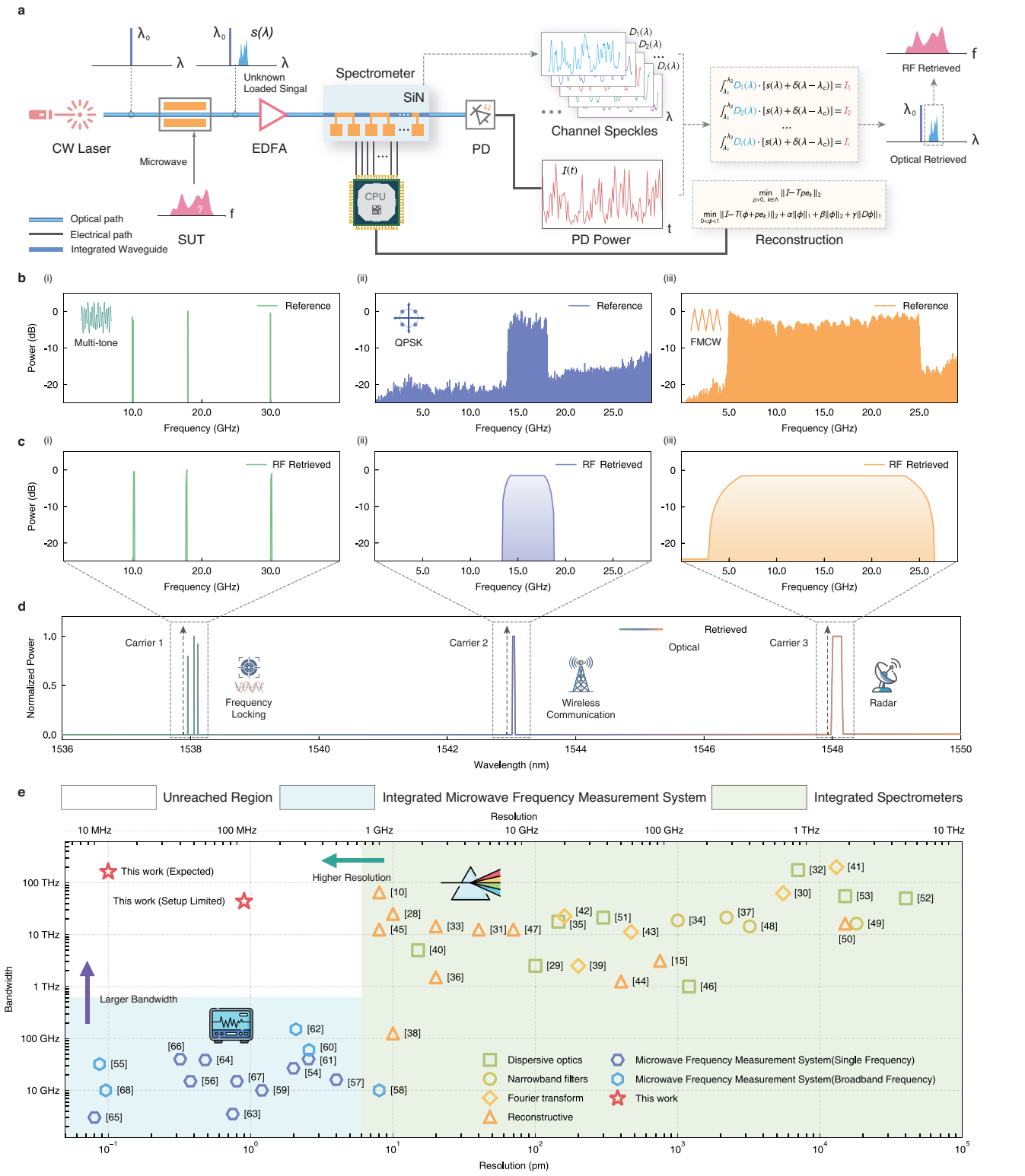


Fig. 5 | Bridging the gap between optical and microwave spectrum analysis. **a** Experimental setup of the HS-PIC microwave analysis system (left) and its reconstruction steps (right). The pink and blue signals correspond to the microwave and optical signals, respectively. **b**–**d** Comparison between the detection system and electrical spectrum analyzer (ESA) for (i) multi-tone, (ii) QPSK, and (iii) FMCW signal recovery. **b** Results measured by ESA. **c** Results obtained from the detection system. **d** Intermediate optical-domain recovery states for different carrier frequencies. **e** A collection of recent works on spectrum analysis shows a gap between optical and microwave detection.

attributed to carrier-induced intensity noise. Consequently, the HS-PIC delivers high-resolution for both microwave and optical spectra and offers a competitive route to unified on-chip frequency domain analysis.

Discussion

Our HS-PIC architecture combines high spectral complexity with extensive tunability. Although the observed super-linear scaling is empirical within the design constraints explored here, it may persist

over a larger circuit area before accumulated loss and thermal tuning noise lead to saturation. Together, this scalable complexity and tunability open opportunities across multiple photonic applications. In optical communications, the tunable high-complexity response can be used to modulate frequency combs, even with free spectral ranges (FSRs) under 100 GHz. It simplifies communication links by removing the requirement for bulky AWGs or MUX/DeMUX components[69]. For spectroscopic sensing, the HS-PIC provides a compact and stable all-optical alternative to hybrid optical-electronic sampling such as dual-comb spectroscopy, which typically suffers from stability issues and integration challenges at sub-GHz resolutions[7,70–72]. Optical computing also benefits from the HS-PIC, which serves as a compact alternative to cascaded micro-ring convolution kernels[8]. This substitution could scale computational density at least 10 times by increasing wavelength channels within a reduced footprint. Future integration with cross-band reflectors and low-loss electro-optic phase modulators may further extend its operational bandwidth and switching speed, enabling hyper-spectral control within a compact device.

Beyond specific applications, the RID framework establishes a scalable design strategy that integrates device physics, system functionality, and design automation. While applied here to optimize the HS-PIC, this methodology naturally extends to broader photonic signal processing platforms. Similar to other gradient-based methods, RID can be influenced by the structural growth sequence and does not guarantee global optimality; instead, it ensures that the resulting design satisfies the prescribed target and fabrication constraints. By converting the traditionally manual design process into an automated workflow, RID brings photonic synthesis closer to electronic design automation (EDA) and may facilitate the development of complex photonic integrated circuits for computing, communication, and sensing.

Methods

Numerical simulation of conventional PIC and HS-PIC

The device performance of multi-mode interferometers and reflective nodes is simulated with three-dimensional Finite-difference time-domain (FDTD) using Ansys Lumerical 2023 R2.2. The system-level simulations are performed using INTERCONNECT with scattering parameters extracted from FDTD. To compare the complexity scaling, we randomly parameterize 100 cascade MZIs, waveguide Bragg gratings and one-dimensional HS-PICs with 2–30 stages. The arm difference (ΔL) of MZI and waveguides in HS-PIC shares the same uniform distribution to guarantee identical time delay.

Recursive inverse design algorithms

The RID method determines all waveguide network parameters through a sequence of decomposition, base-case solution, and unwinding stages. Decomposition was implemented in Python using the transfer matrix method (TMM). The process initiates with a fundamental two-component structure with randomly initialized waveguide lengths. Additional reflective nodes are then sequentially added to the optical link. At each step, we utilized PSO to identify the optimal insertion position and connecting waveguide length for the new device, aiming to maximize the decorrelation capability of the transmission spectrum. At each iteration, the spectral complexity C_n of the optimized n-stage circuit is evaluated; the recursion terminates as soon as $C_n \geq C$, and the converged value of n is reported as the node count N. The FOM also incorporates constraints on the correlation function side-lobes to prevent periodic spectral responses. This procedure outputs an optimized circuit architecture that defines the position, reflectivity, and interconnection length for every component. The complete design parameters are presented in Supplementary Tables 4, 5.

The geometry inverse design is performed using the open-source toolkit Meep. The design region is set to 2 × 13 μm and parameterized as a matrix with 40 grids per micrometer. Geometry constraints of 150–350 nm are applied based on the target wavelength to ensure the fabrication compatibility. It works by applying erosions and dilations to the design and verifying the consistency of the resulting topology through an indicator function. The optimization procedure consisted of two stages: a continuous optimization phase followed by binarization. Device structure parameters were optimized from an arbitrary initial constant value to the final design after 300 iterations. For different nodes, distinct source wavelengths and target figures of merit were specified to meet the requirements of various HS-PIC configurations. Supplementary Fig. 10 presents a representative optimization process, Supplementary Figs. 11, 12 show the optimized-device field distributions, and Supplementary Fig. 13 shows SEM images of all optimized devices.

The unwinding stage implements the same recursive strategy used in the decomposition phase. In this step, the ideal transmission matrices of the reflective nodes are sequentially replaced by simulated matrices representing the actual physical structures. To compensate for mismatches between the physical and ideal responses, a particle swarm optimization (PSO) algorithm fine-tunes the waveguide lengths within a limited range. For high-complexity targets, the objective function remains consistent with the decomposition stage; however, for response shaping, the figure of merit is redefined as $\mathrm{FoM} = \sqrt{\sum_{n\in D}(H(n) - T(n))^2}$. As detailed in Supplementary Note 5, we demonstrated spectral shaping using single-point, flat-top, W-shaped, and composite responses. The analysis indicates that the HS-PIC achieves diverse response profiles by modifying only 25% of the waveguide length, which confirms its suitability for narrowband filtering applications.

Device fabrication

The HS-PICs were built on a silicon nitride-on-insulator wafer with a 0.3-μm-thick Si_3N_4 device layer deposited by low-pressure chemical vapor deposition (LPCVD) and a wet thermal buried oxide layer. The waveguide patterns and marks for backend processes were defined by electron-beam lithography (EBL) and transferred onto the Si_3N_4 layer by inductively coupled plasma reactive ion etching (ICP-RIE). As illustrated in Supplementary Fig. 13, we employed SEM to characterize the morphological features of the reflectors across the spectral range from 780 nm to 2000 nm, showing consistency with the design. Then, SiO_2 cladding was deposited by the Plasma Enhanced Chemical Vapor Deposition (PECVD). The electrodes (80 nm Tungsten) were then defined by ultraviolet lithography, followed by magnetron sputtering and a lift-off process. Finally, the waveguide couplers were polished by a cross-sectioning device to ensure low optical loss.

Experimental setup of spectrometer

Supplementary Fig. 16 displays the experimental setup of the spectrometer. During calibration, a broadband amplified spontaneous emission source (ASE, MC Fiber Optics, MCASE-CL-20-T-2-1-FA-M2) is injected into the spectrometer chip. Voltage control sequences are applied to the chip via a custom MCU control board, while a commercial benchtop optical spectrum analyzer (Yokogawa AQ6370D) records the spectral response of each channel. We calibrated the spectrometer with three distinct ASE light sources, separately centered at 1310 nm, 1550 nm, and 2000 nm. The operational bandwidths of these sources were 40 nm, 75 nm, and 120 nm, respectively. The insufficient bandwidth of the ASE sources is the primary bottleneck that limits the working bandwidth of our spectrometer.

For performance testing, the light source is switched to an unknown input signal. The control system then scans all operating channels in the same sequence as calibration, while a photodiode (PD) monitors the output power in real time. The custom control system is driven by a microcontroller unit (MCU, STM32F103C8T6). It generates

10 parallel control signals via digital-to-analog converters (DAC, AD5328), which are subsequently amplified by operational amplifiers to drive the phase shifter of the spectrometer chip. This architecture enables rapid switching of channels.

In our measurements, the channel switching time is less than 0.5 ms, allowing 1024 channels to be scanned within 0.6 s (1 s was used in our experiments). By updating firmware from a computer, we can flexibly adjust the switching rate, output bit depth, and voltage range, thereby adapting the spectrometer system to a broad spectrum of application scenarios.

Data availability

The data that support the plots within this paper and other findings of this study are available within the article and its supplementary information. The spectrometer acquisition data and reconstruction results are available on GitHub at https://github.com/H2ydrogen/HSPIC_spectrometer. Source data are provided with this paper.

Code availability

The codes that support the findings of this study are available publicly. The code for the spectrometer reconstruction algorithm is available on GitHub at https://github.com/H2ydrogen/HSPIC_spectrometer.

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Author contributions

The concept of this work was conceived by H.H. The HS-PIC theory and RID design were developed by H.H., L.C., and Z.T. The chip was fabricated by Y.W., H.H., H.Z., and Q.Y. The experiments were performed by H.H. and Z.T., with the assistance of C.F., Y.Z., Y.C., R.Y., L.Z., and J.W. The reconstruction algorithms were developed by H.H. and Z.T. All authors participated in writing the paper. The project was under the supervision of L.C.

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Competing interests

The authors declare no competing interests.

Additional information

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